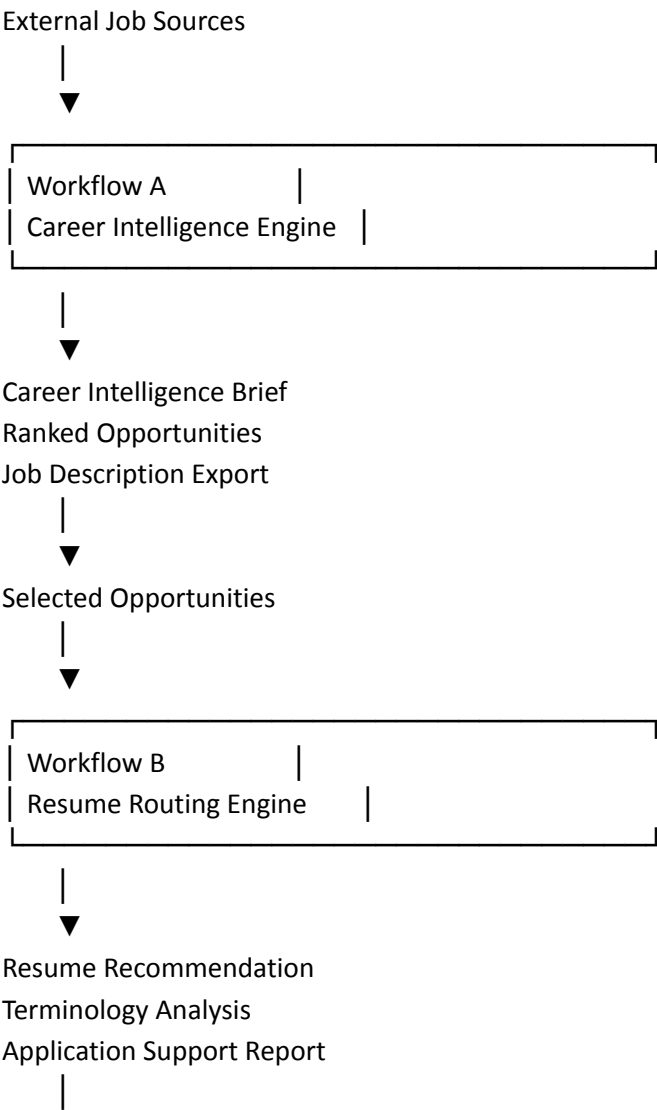


Career Operations Intelligence Platform

Full System Architecture

Version 3.1

System Topology





Human Review
Application Decision

Technology Stack

Infrastructure

- Windows 11
- Docker Desktop
- n8n

External Services

- Gmail API
- Google Gemini 2.5 Pro
- Obsidian Local REST API

Local AI Services

- Ollama
- Qwen 2.5 14B

Runtime Languages

Workflow A

- JavaScript
- n8n Expressions

Workflow B

- Python 3.13

Data Formats

- TXT
 - JSON
 - Markdown
-

Workflow A Architecture

Data Sources

Current sources include: -

- Glassdoor Alerts
- Indeed Alerts
- Adzuna Alerts
- Company Career Alerts
- Recruiter Outreach
- Manual Job Discovery

All opportunity signals arrive through Gmail.

Intake Layer

Gmail Retrieval

Query:

newer_than:1d

Retrieved fields:

```
{  
  "subject": "",  
  "from": "",  
  "date": "",  
  "body": ""  
}
```

Purpose:

Retrieve all opportunity-related messages generated during the previous 24 hours.

Preprocessing Layer

Objective

Reduce noise before AI processing.

Subject Filtering

Ignore categories:

- application confirmations
- password resets
- billing notices
- account notifications
- onboarding messages

Example:

```
[  
  "application received",  
  "application submitted",  
  "password reset",  
  "subscription"  
]
```

URL Preservation

URLs are extracted before text cleanup.

Regex:

```
/https?:\\\/[^\s<>"]+/g
```

Output:

```
{
```

```
"links": [],  
"linkCount": 0  
}
```

Text Cleanup

Removes:

- forwarded email wrappers
- quoted replies
- signature blocks
- excess whitespace

Normalization:

```
text  
 .replace(/\n{3,}/g, "\n\n")  
 .replace(/\s{2,}/g, " ")
```

Token limit:

```
text.slice(0,3000)
```

Aggregation Layer

All normalized emails are assembled into a single payload.

Output:

```
{  
  "emails": [...]  
}
```

Purpose:

Provide Gemini with the complete daily opportunity set in a single inference context.

Gemini Pass 1

Canonical Job Extraction

Purpose:

Identify opportunities without evaluating them.

Core Instruction:

You are an information extraction system.

Responsibilities:

- identify jobs
- split digest emails
- preserve source metadata
- collect candidate URLs

Explicitly prohibited:

- ranking
- scoring
- recommendations
- resume evaluation

Output:

```
{
  "jobs": [
    {
      "title": "",
      "company": "",
      "location": "",
      "description": "",
      "candidate_urls": [],
      "source_email_id": "",
      "confidence": 0,
```

```
"flags": []  
}  
]  
}
```

Canonicalization Layer

Implemented through JavaScript transformation nodes.

Responsibilities:

- deduplication
- schema normalization
- URL preservation
- metadata consolidation

Job identity generation:

```
company  
+  
title  
+  
location  
+  
url
```

Output:

```
{  
  "jobs": [...]  
}
```

Canonicalization exists to prevent duplicate opportunities from propagating into evaluation stages.

Gemini Pass 2

Career Intelligence Evaluation

Purpose:

Determine which opportunities deserve attention.

Input:

```
{  
  "jobs": [...]  
}
```

Evaluation dimensions:

- Interview Probability
 - Career Capital
 - Long-Term Optionality
 - Organizational Development Value
 - Geographic Practicality
 - Compensation
-

Recommendation Framework

90+ PRIORITY APPLY

80-89 APPLY

70-79 REVIEW

<70 SKIP

Output:

```
{  
  "rank": 1,  
  "title": "",  
  "company": "",  
  "fit_score": 0,  
  "recommendation": "",  
}
```



```
"confidence": 0.0,  
"trajectory_alignment": "",  
"why_good_fit": "",  
"career_capital": "",  
"institutional_training_value": "",  
"risks": ""  
}
```

Reporting Layer

Generates Markdown reports.

Output destination:

Obsidian Vault/
Career Intelligence/

Generated sections:

- Workflow Metrics
 - Top Opportunity
 - Ranked Opportunities
 - Career Capital Assessment
 - Recommendation Summary
 - Risk Summary
-

Workflow A Observability

Metrics tracked:

```
{  
  "emails_processed": 0,  
  "jobs_extracted": 0,  
  "duplicates_removed": 0,  
  "jobs_reviewed": 0,
```

```
"priority_apply": 0,  
"apply": 0,  
"review": 0,  
"skip": 0,  
"average_confidence": 0.00  
}
```

Purpose:

- workflow monitoring
 - debugging
 - trend analysis
 - quality control
-

Workflow B Architecture

Purpose

Workflow B determines how a selected opportunity should be approached.

Workflow B answers:

Which resume should be used?

What terminology may warrant review?

Repository Structure

Qwen_Resume_Workflow/

config/
resumes/
input_jobs/
output/
temp/

scripts/
run_all.py

Input Layer

Workflow A exports selected opportunities as text files.

input_jobs/YYYY-MM-DD/

Example:

01_Company_Role.txt
02_Company_Role.txt
03_Company_Role.txt

Resume Profile Repository

Location:

resumes/resume_profiles.json

Resume metadata includes:

```
{  
  "resume_id": "",  
  "primary_type": "",  
  "focus_areas": [],  
  "skills": []  
}
```

The system evaluates resume metadata rather than editing resume content.

Preprocessing Layer

Script:

01_preprocess.py

Responsibilities:

- locate newest job folder
- load job descriptions
- load resume profiles
- normalize text
- construct runtime payload

Output:

temp/runtime_payload.json

Qwen Routing Layer

Script:

02_qwen_call.py

Model:

qwen2.5:14b

Endpoint:

<http://localhost:11434/api/generate>

Purpose:

- resume selection
- terminology analysis

Output:

```
{  
  "company": "",  
  "title": "",  
  "recommended_resume": "",  
  "confidence": 0.00,  
  "skills_additions": [],  
  "skills_removals": [],  
  "flags": []  
}
```

Raw responses preserved:

temp/qwen_raw_output.txt

Parsed responses:

temp/qwen_output.json

Report Generation Layer

Script:

03_report.py

Outputs:

Resume_Routing_Report.json

Resume_Routing_Report.md

Purpose:

Generate human-readable application-support reports.

Confidence & Flagging

Every recommendation includes:

```
{  
  "confidence": 0.88,  
  "flags": []  
}
```

Possible flags:

unclear_job_description
weak_resume_match
multiple_role_types
keyword_sparse

Purpose:

Expose uncertainty rather than hide it.

Reliability Controls

Several controls exist to mitigate common LLM failure modes.

Implemented controls include:

- structured JSON outputs
- schema validation
- canonicalization layers
- deterministic preprocessing
- confidence scoring
- raw output preservation
- provenance tracking
- workflow separation
- human review checkpoints

These controls were introduced in response to observed issues including hallucination, inconsistent scoring, schema drift, metadata loss, and duplicate opportunity generation.

Human Control Boundary

The platform may:

- discover opportunities
- extract information
- evaluate opportunities
- rank opportunities
- recommend resumes
- identify terminology

The platform may not:

- submit applications
- rewrite resumes
- fabricate qualifications
- alter employment history
- make final career decisions

All consequential actions remain under human control.

End State

The Career Operations Intelligence Platform functions as a layered career decision-support system composed of workflow automation, deterministic processing, and AI-assisted evaluation.

Workflow A performs strategic opportunity intelligence.

Workflow B performs tactical resume routing and terminology analysis.

Together they provide a structured process for transforming large volumes of unstructured job-search information into explainable, reviewable, and actionable career intelligence while maintaining human oversight throughout the decision process.

